Predicting 30-Day Hospital Readmission for Patients with Diabetes

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Contents

Introduction **1**

Question 1: Which factors best predict readmission within 30 days? **1**
- Introduction . . . 1
- Approach . . . 2
- Data analysis and results . . . 2
- Discussion . . . 3

Question 2: Which combinations of characteristics are associated with early readmission? **3**
- Introduction . . . 3
- Approach . . . 3
- Data analysis and results . . . 3
- Discussion . . . 4

References **5**

Introduction

Hospital readmission within 30 days of discharge is costly, often preventable, and used in the United States as a measure of hospital quality. Patients with diabetes are readmitted more often than most, because diabetes complicates recovery from almost any illness. This report uses the **Diabetes 130-US Hospitals for Years 1999–2008** dataset (Strack et al., 2014; UCI Machine Learning Repository, CC BY 4.0), which records 101,766 inpatient stays of patients with diabetes at 130 US hospitals. Each stay is described by the patient's age band, gender and race; the admission type and source and the discharge destination; the length of stay and numbers of lab tests, procedures, medications and diagnoses; the patient's outpatient, emergency and inpatient visits in the previous year; up to three ICD-9 diagnosis codes; two glucose tests (HbA1c and serum glucose); and the status of 23 diabetes medications. The outcome records whether the patient was readmitted within 30 days, after 30 days, or not at all.

To prepare the data, stays ending in death or hospice care were removed (those patients cannot be readmitted), and only each patient's first stay was kept, so that every patient counts once. `weight` (97% missing) and the payer code were dropped, missing race and medical specialty became their own categories, the glucose tests' "not performed" value was kept as a category, and the 700+ diagnosis codes were grouped into nine clinical categories. The cleaned data contain **69,987 patients** and no missing values; **8.98%** were readmitted within 30 days. All analysis was done twice, in Python and in R, with matching results.

Question 1: Which factors best predict readmission within 30 days?

Introduction

If a hospital could tell at discharge which patients are likely to return within a month, it could target follow-up calls, medication reviews or home visits at them. This question asks how well the recorded patient and

hospital-stay information predicts 30-day readmission, and which factors drive the prediction. It uses 25 predictors (demographics, admission and discharge details, stay measures, prior-year visits, diagnosis groups, glucose tests and diabetes medications) and the binary outcome *readmitted within 30 days* (yes or no). I am interested in it because it turns a large administrative dataset into something a care team could act on.

Approach

Three classifiers were compared. **Logistic regression** is a transparent baseline whose coefficients, expressed as odds ratios, show each factor's direction and size of effect. A **decision tree** (depth 6, at least 100 patients per leaf) gives readable rules. A **random forest** (300 trees) captures interactions and non-linear effects. Numeric features were standardized (z-scores) so coefficients are comparable, and the three heavily right-skewed visit counts were log-transformed first. Categorical features were one-hot encoded with a reference level. Because only 9% of patients are readmitted, a model predicting "no" for everyone would be 91% accurate and useless, so each model weighted the classes equally and was judged by **ROC-AUC** (how well it ranks readmitted patients above others), **recall** (share of readmissions flagged) and **precision** (share of flagged patients actually readmitted). The data were split 80/20, stratified on the outcome, with 5-fold cross-validation on the training set. Feature importance was measured by odds ratios and by **permutation importance** (the drop in test AUC when one feature is shuffled).

Data analysis and results

Table 1: Test-set performance (Python, threshold 0.5). The last column shows the R replication.

Model	CV ROC-AUC	Test ROC-AUC	Recall	Precision	Share flagged	Test ROC-AUC (R)
Logistic regression	0.639 ± 0.004	0.640	0.539	0.135	0.359	0.642
Decision tree	0.627 ± 0.006	0.634	0.567	0.129	0.395	0.620
Random forest	0.646 ± 0.007	**0.647**	0.348	**0.165**	0.190	0.647

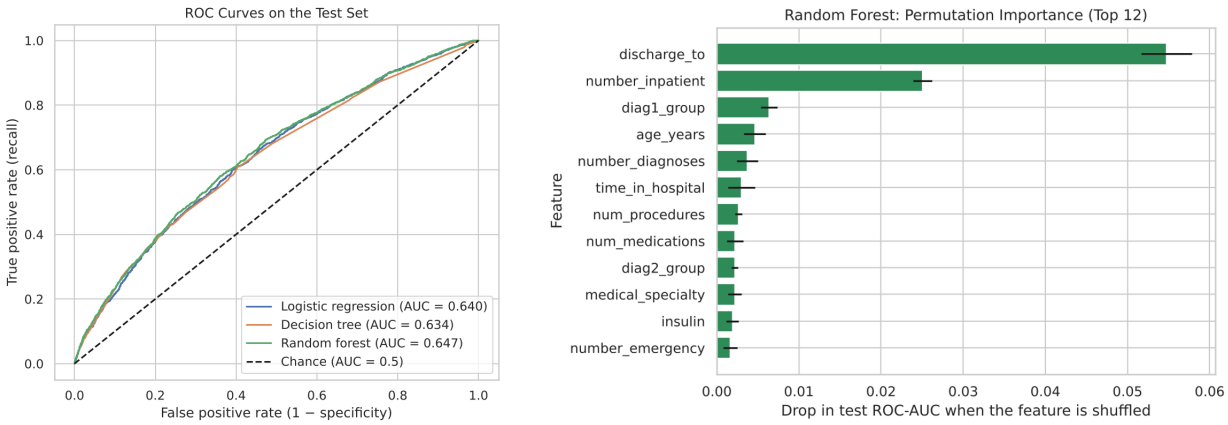


Figure 1: ROC curves on the test set (left) and the random forest's permutation importance (right).

All three models predict better than chance, with test ROC-AUC of 0.63–0.65 and stable cross-validated scores (Table 1, Figure 1). The random forest ranks best but gains little over logistic regression. Two features dominate: **discharge destination** (shuffling it lowers the forest's AUC by 0.055, more than twice any other feature) and **prior-year inpatient visits** (0.025). In the logistic regression, patients discharged home have **0.45 times the odds** of readmission of those discharged to a care facility, and each standard deviation of log inpatient visits raises the odds by 24% (odds ratio 1.24). Smaller effects come from age, length of stay,

number of diagnoses and emergency visits (each about 7% per standard deviation), being on any diabetes medication (1.28) and nephrology as the admitting specialty (1.70 relative to cardiology).

Discussion

Early readmission is driven mainly by **how sick the patient is at discharge** (patients who need a care facility return far more often) and by **how often they have been hospitalized before**. Other factors add only a little. The models are only moderately accurate: the best ranks a readmitted patient above a non-readmitted one about 65% of the time, and the random forest, which flags 19% of patients, catches 35% of readmissions with a precision of 16.5%, almost double the 9% base rate. That is useful for prioritizing follow-up but misses most readmissions. The more flexible random forest barely beats logistic regression, which suggests the limit lies in the data rather than the model: much of what causes readmission happens after discharge (medication adherence, follow-up care, home support) and is not recorded. The data are also from 1999–2008, and a hospital would tune the decision threshold to its own capacity.

Question 2: Which combinations of characteristics are associated with early readmission?

Introduction

Question 1 ranks single factors. Clinicians, however, think in profiles: an older patient admitted through the emergency room and discharged to a nursing facility. This question looks for **combinations** of patient characteristics, stay details and treatments that occur together with 30-day readmission, stated as readable rules with the readmission rate of each group. It uses the same patients and 15 of the same attributes (with numeric ones binned) plus the outcome. I am interested in it because rules like these are easy to communicate and could become simple screening criteria.

Approach

Association rule mining with the **Apriori** algorithm was used. Each patient became a "basket" of 16 items, one per attribute (for example `discharge=Home` or `prior_inpatient=2+`); numeric features were discretized into clinically sensible bins, because Apriori needs categorical items. For a rule A $\rightarrow$ {readmitted within 30 days}, **support** is the share of all patients who have A *and* were readmitted, **confidence** is the readmission rate among patients with A, and **lift** is confidence divided by the overall 8.98% rate. A **minimum support of 0.5%** ensured each rule rests on at least 350 patients, and rules had at most four items. Rules with **lift** $\geq$ **1.5** mark higher-risk groups and **lift** $\leq$ **0.67** lower-risk groups. Because Apriori produces many near-duplicate rules, a rule was kept only if its confidence beat every shorter rule contained in it (**redundancy pruning**).

Data analysis and results

Apriori found 73,622 frequent itemsets and 2,811 rules predicting readmission; after pruning, **77 higher-risk** and **65 lower-risk** rules remained. Python (mlxtend) and R (arules) produced identical rules.

Table 2: Selected non-redundant rules with consequent {readmitted within 30 days}.

Rule (left-hand side)	Patients	Readmission rate (confidence)	Lift
{HbA1c not tested, prior inpatient stays 2+}	2,077	22.1%	2.46
{prior inpatient stays 2+}	2,397	21.5%	2.39
{on diabetes medication, circulatory primary diagnosis, discharged to care facility}	2,249	18.5%	2.06
{emergency admission, age 60–79, discharged to care facility}	2,842	18.2%	2.03

Rule (left-hand side)	Patients	Readmission rate (confidence)	Lift
{age 60–79, on diabetes medication, discharged to care facility}	4,394	17.9%	1.99
{age under 60, discharged home, no prior inpatient stay}	16,636	5.3%	0.59
{age under 60, fewer than 7 diagnoses, no prior inpatient stay}	9,940	5.0%	0.56

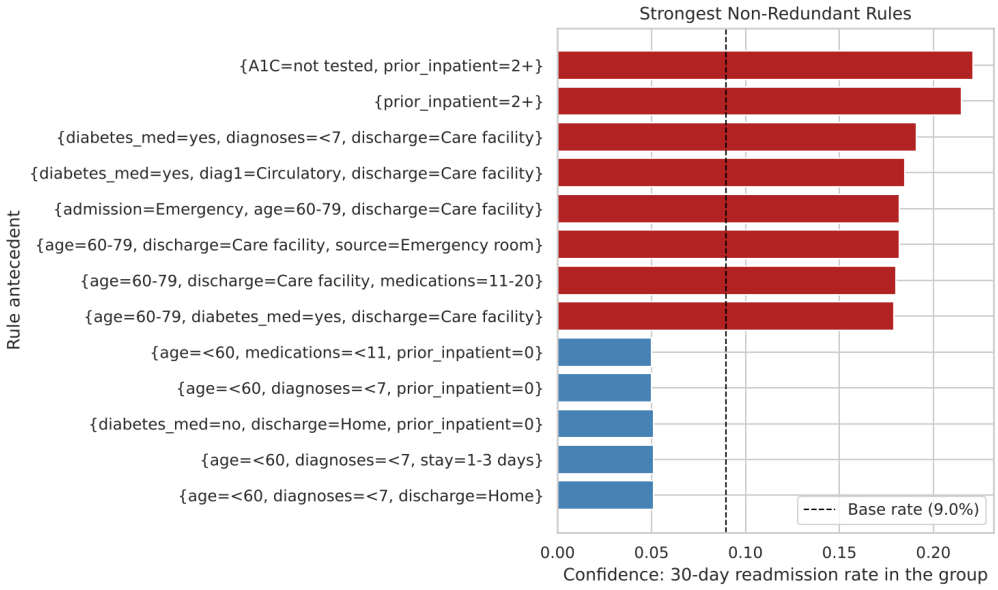


Figure 2: Readmission rate of the strongest non-redundant rules; the dashed line is the 9.0% base rate.

The higher-risk rules fall into two families. Patients with **two or more inpatient stays in the previous year** are readmitted at 21.5%, 2.4 times the average, and adding *HbA1c not tested* raises this only slightly, to 22.1%. Patients **discharged to a care facility**, particularly those aged 60–79, admitted as emergencies or with a circulatory primary diagnosis, are readmitted at 17–19%, about twice the average. At the other end, large groups of **younger patients with no prior stay**, especially those discharged home with few diagnoses or medications, are readmitted at only about 5% (Table 2, Figure 2).

Discussion

The rules confirm the two strongest predictors from Question 1 and turn them into concrete, actionable groups: a patient with two or more hospital stays in the past year has better than a 1-in-5 chance of returning within a month, and an older patient discharged to a care facility after an emergency admission has nearly a 1-in-5 chance. They also show where risk concentrates, for example that care-facility risk is highest among patients aged 60–79 and emergency admissions, which a single odds ratio does not reveal. Some rules are counter-intuitive, such as care-facility patients with *fewer* recorded diagnoses appearing at higher risk; this may reflect how diagnoses were coded for transferred patients, a reminder that association rules describe co-occurrence, not cause. The results also depend on the chosen bins and the 0.5% minimum support, which hides smaller but more extreme groups (patients with three or more prior stays are readmitted at 26.5% but appear only inside the “2+” item).

References

Strack, B., DeShazo, J. P., Gennings, C., Olmo, J. L., Ventura, S., Cios, K. J., & Clore, J. N. (2014). Impact of HbA1c measurement on hospital readmission rates: Analysis of 70,000 clinical database patient records. *BioMed Research International*, 2014, 781670.

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Code: `python/01-03_*.ipynb` and `r/01-03_*.Rmd` in the project folder reproduce every result.